

A counterexample to an aggregate ϕ - η mixing comparison

Autonomous solution packet for arXiv:0711.0986

11 August 2026

Abstract

Kontorovich conjectured that a sum of finite-sequence ϕ -mixing coefficients is controlled by one plus the maximal row sum of the η -mixing matrix. We give a full-support counterexample on a finite alphabet. It consists of independent weak binary channels whose outputs are delayed by a common distance. Every nonzero η row has sum one, but the weak channels accumulate in the ϕ coefficients. An elementary likelihood-ratio estimate makes the conjectured left side exceed 2.933 while the right side equals 2.

1 Source conjecture and definitions

The source paper is L. Kontorovich, *Constructing processes with prescribed mixing coefficients* [1]. It first appeared as arXiv:0711.0986. Page 2 defines $\bar{\eta}_{ij}$, displays the conjecture, and says that it remains open. The page is reproduced next.

be the distribution of $X_j^n \equiv (X_j, \dots, X_n)$ conditioned on $X_1^i = x$. For $y \in \Omega^{i-1}$ and $w, w' \in \Omega$, define

$$\eta_{ij}(y, w, w') = \left\| \mathcal{L}(X_j^n | X_1^i = yw) - \mathcal{L}(X_j^n | X_1^i = yw') \right\|_{\text{TV}}, \quad (1)$$

where $\|\cdot\|_{\text{TV}} \equiv \frac{1}{2} \|\cdot\|_1$ is the total variation norm; likewise, define

$$\bar{\eta}_{ij} = \max_{y \in \Omega^{i-1}, w, w' \in \Omega} \eta_{ij}(y, w, w'). \quad (2)$$

This notion of mixing is by no means new; it can be traced (at least implicitly) to Marton's work [16] and is quite explicit in Samson [17] and Chazottes et al. [7]. We are not aware of a standardized term for this type of mixing, and have referred to it as η -mixing in previous work [13]. It was observed in [17] that the ϕ -mixing coefficients bound the η -mixing ones:

$$\bar{\eta}_{ij} \leq 2\phi_{j-i},$$

and conjectured in [11] that

$$\frac{1}{2} \sum_{i=1}^{n-1} \phi_i \leq 1 + \max_{1 \leq i < n} \left[\sum_{j=i+1}^n \bar{\eta}_{ij} \right];$$

the latter remains open.

In all instances, η -mixing has come up in the context of concentration of measure. In particular, define Γ and Δ to be upper-triangular $n \times n$ matrices, with $\Gamma_{ii} = \Delta_{ii} = 1$ and

$$\Gamma_{ij} = \sqrt{\bar{\eta}_{ij}}, \quad \Delta_{ij} = \bar{\eta}_{ij} \quad (3)$$

for $1 \leq i < j \leq n$.

Samson [17] proved that any distribution μ on $[0, 1]^n$ and any convex $f : [0, 1]^n \rightarrow \mathbb{R}$ with $\|f\|_{\text{Lip}} \leq 1$ (with respect to ℓ_2) satisfy

$$\mu\{|f - \mu f| > t\} \leq 2 \exp\left(-\frac{t^2}{2\|\Gamma\|_2^2}\right) \quad (4)$$

where $\|\Gamma\|_2$ is the ℓ_2 operator norm.

Chazottes et al. [7] and independently, the author with K. Ramanan [13] showed that any distribution μ on Ω^n and any $f : \Omega^n \rightarrow \mathbb{R}$ with $\|f\|_{\text{Lip}} \leq n^{-1/2}$ (with respect to the Hamming metric) satisfy

$$\mu\{|f - \mu f| > t\} \leq 2 \exp\left(-\frac{t^2}{2\|\Delta\|_\infty^2}\right) \quad (5)$$

where $\|\Delta\|_\infty$ is the ℓ_∞ operator norm ($\|\Delta\|_\infty$ may be replaced by $\|\Delta\|_2$ and [7] achieves a better constant in the exponent).

Results of type (4) and (5) are known as *concentration of measure* inequalities; broadly, they assert that any “sufficiently continuous” function is tightly concentrated about its mean. Such bounds have a remarkable range of applications, spanning abstract fields such as asymptotic Banach

Figure 1: The source definition and conjecture on [1, p. 2].

Let $X_1, \dots, X_n$ be a random sequence. For $1 \leq i < j \leq n$, the source coefficient is

$$\bar{\eta}_{ij} = \max_{y, w, w'} \text{TV}(\mathcal{L}(X_j^n \mid X_1^i = yw), \mathcal{L}(X_j^n \mid X_1^i = yw')) ,$$

where total variation has range $[0, 1]$. Our law will have full support, so all atoms in this maximum have positive probability. We use the standard finite-sequence ϕ coefficient

$$\phi_r = \max_{1 \leq t \leq n-r} \sup_{\substack{A \in \sigma(X_1^t), \mathbb{P}(A) > 0 \\ B \in \sigma(X_{t+r}^n)}} |\mathbb{P}(B \mid A) - \mathbb{P}(B)| . \quad (1)$$

The conjecture displayed in the source is

$$\frac{1}{2} \sum_{r=1}^{n-1} \phi_r \leq 1 + \max_{1 \leq i < n} \sum_{j=i+1}^n \bar{\eta}_{ij} . \quad (2)$$

2 Proof intuition

Put a fair latent sign at time k and reveal just one slightly correlated copy at time $k + L$. The resulting k th η row is a flat block of height c and length L . Choosing $c = 1/L$ makes every row sum one. Now place L such channels independently in parallel. At a middle cut, the past reveals many latent signs and the future contains their noisy copies. Conditioning all the latent signs to be positive changes a product of m fair signs into a product of m signs of bias c . That total variation is of order $c\sqrt{m}$. Summing over the L lags gives order $cL^{3/2} = \sqrt{L}$, which cannot be controlled by the fixed row sum.

3 Counterexample

Theorem 1. *The comparison (2) is false, even for a full-support probability measure on a finite alphabet. More precisely, there is such a sequence of length $n = 2048$ for which*

$$\max_i \sum_{j=i+1}^n \bar{\eta}_{ij} = 1 \quad \text{but} \quad \frac{1}{2} \sum_{r=1}^{n-1} \phi_r > 2.933 .$$

Thus the right side of (2) is 2 and the inequality is violated.

Proof. We first give a construction for an integer $L \geq 2$ and a parameter $0 < c < 1$, with $n = 2L$. For each $1 \leq k \leq L$, let (Z_k, Y_k) be an independent pair of signs with

$$\mathbb{P}(Z_k = 1) = \mathbb{P}(Z_k = -1) = \frac{1}{2}, \quad \mathbb{P}(Y_k = Z_k \mid Z_k) = \frac{1+c}{2} .$$

The pairs are mutually independent. Also take mutually independent fair signs $U_{k,t}$, independent of all pairs, for $1 \leq k \leq L$ and $t \notin \{k, k+L\}$. Define

$$A_t^{(k)} = \begin{cases} Z_k, & t = k, \\ Y_k, & t = k+L, \\ U_{k,t}, & t \notin \{k, k+L\}, \end{cases} \quad X_t = (A_t^{(1)}, \dots, A_t^{(L)}) \in \{-1, 1\}^L .$$

Because $0 < c < 1$, all four outcomes of every pair (Z_k, Y_k) have positive probability. Hence the law of X_1^n has full support.

We calculate its η matrix. If $i \leq L$, the only coordinate of X_i that can affect a future variable is Z_i , and its only dependent future variable is Y_i at time $i + L$. Conditional on $Z_i = 1$ and $Z_i = -1$, the two laws of Y_i have total variation c . All other tail coordinates have the same conditional law. If $i > L$, no coordinate of X_i has a dependent future variable. It follows exactly that

$$\bar{\eta}_{ij} = \begin{cases} c, & 1 \leq i \leq L \text{ and } i < j \leq i + L, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Consequently the maximal row sum is Lc .

We next lower-bound ϕ_r for $1 \leq r \leq L$. Use the cut $t = L$ in (1) and condition on the positive-probability past event

$$E_r = \{Z_r = Z_{r+1} = \dots = Z_L = 1\}.$$

The future X_{L+r}^{2L} contains $Y_r, \dots, Y_L$. Marginally these are $m = L - r + 1$ independent fair signs, while conditional on E_r they are independent signs of mean c . Taking the supremum over future events gives

$$\phi_r \geq d_m(c) := \text{TV}(\text{Rad}(c)^{\otimes m}, \text{Rad}(0)^{\otimes m}), \quad m = L - r + 1, \quad (4)$$

where $\text{Rad}(c)$ denotes a sign of mean c .

It remains to estimate $d_m(c)$ from below. Under the fair product law Q , write the coordinate signs as $\varepsilon_1, \dots, \varepsilon_m$ and put $S = \sum_a \varepsilon_a$. The likelihood ratio of the biased product law is

$$\Lambda = \prod_{a=1}^m (1 + c\varepsilon_a) = 1 + cS + R.$$

Orthogonality of the Rademacher monomials gives

$$\mathbb{E}_Q R^2 = (1 + c^2)^m - 1 - mc^2. \quad (5)$$

If $mc^2 \leq 1$ and $z = mc^2$, then

$$(1 + c^2)^m - 1 - mc^2 \leq e^z - 1 - z \leq \frac{1}{2}e^z z^2,$$

so

$$\mathbb{E}_Q |R| \leq \frac{e^{1/2}}{\sqrt{2}} mc^2. \quad (6)$$

For completeness, $\mathbb{E}_Q S^2 = m$ and $\mathbb{E}_Q S^4 = 3m^2 - 2m$. Hölder interpolation gives

$$\mathbb{E}_Q |S| \geq \frac{(\mathbb{E}_Q S^2)^{3/2}}{(\mathbb{E}_Q S^4)^{1/2}} \geq \sqrt{\frac{m}{3}}.$$

Since $d_m(c) = \frac{1}{2} \mathbb{E}_Q |\Lambda - 1|$, the triangle inequality and (6) yield

$$d_m(c) \geq \frac{c}{2} \sqrt{\frac{m}{3}} - \frac{e^{1/2}}{2\sqrt{2}} mc^2. \quad (7)$$

Finally choose $c = 1/L$. Then $mc^2 \leq 1$, and (3) gives maximal row sum $Lc = 1$. Using (4), (7), $\sum_{m=1}^L \sqrt{m} \geq \frac{2}{3}L^{3/2}$, and $\sum_{m=1}^L m = L(L+1)/2$, we obtain

$$\begin{aligned} \frac{1}{2} \sum_{r=1}^{2L-1} \phi_r &\geq \frac{1}{2} \sum_{m=1}^L d_m(1/L) \\ &\geq \frac{\sqrt{L}}{6\sqrt{3}} - \frac{e^{1/2}(L+1)}{8\sqrt{2}L}. \end{aligned}$$

For $L = 1024$, this last quantity is $2.9333313745\dots > 2$. The right side of (2) is $1 + 1 = 2$, proving the claim. $\square$

4 Verification and scope

The script `code/check_counterexample.py` evaluates both the analytic bound and the exact binomial total variations for all $1 \leq m \leq 1024$. It reports analytic left-side lower bound $2.9333313745\dots$ and exact product lower value $4.2591246806\dots$, against right side 2. The exact evaluation is a check, not a dependency of the proof.

A bounded novelty search on 11 August 2026 covered the run registry, the exact paper title, the phrases “eta-mixing”, “phi-mixing”, and “the latter remains open”, and the displayed comparison. It found the source and bibliographic copies but no later primary proof or counterexample. A broader primary-source query timed out, and no complete citation-graph sweep was performed, so novelty confidence is moderate. Specialist review should focus on agreement with the source’s finite-sequence convention for ϕ_r and on the exact calculation (3).

References

- [1] L. Kontorovich, *Constructing processes with prescribed mixing coefficients*, Statistics & Probability Letters 78 (2008), 2910–2915; arXiv:0711.0986; doi:10.1016/j.spl.2008.04.016.